

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/08/2020

Quiz 9: Section 4.1, 4.2, 4.3

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Section 4.1

3 pts

1. Find the absolute max/min **values** of $f(x) = x^4 - 2x^3$ on the interval $[-2, 2]$.

- (A) The absolute max. of $f(x)$ on $[-2, 2]$ is 32 and the absolute min. is 0.
- (B) The absolute max. of $f(x)$ on $[-2, 2]$ is $3/2$ and the absolute min. is $-27/16$.
- (C) The absolute max. of $f(x)$ on $[-2, 2]$ is 32 and the absolute min. is $-27/16$.**
- (D) The absolute max. of $f(x)$ on $[-2, 2]$ is $3/2$ and the absolute min. is -2 .

Because f is a polynomial, its derivative exists everywhere. So if f has critical points, they are points at which $f'(x) = 0$. Computing f' and setting it equal to zero, we have

$$f'(x) = 4x^3 - 6x^2 = 2x^2(2x - 3) = 0.$$

Solving this equation gives the critical points $x = 0$ and $x = 3/2$, both of which lie in the interval $[-2, 2]$; these points and the endpoints are candidates for the location of absolute extrema. Evaluating f at each of these points, we have

$$f(-2) = 32, \quad f(0) = 0, \quad f(3/2) = -27/16, \quad \text{and} \quad f(2) = 0.$$

The largest of these function values is $f(-2) = 32$, which is the absolute maximum of f on $[-2, 2]$. The smallest of these values is $f(3/2) = -27/16$, which is the absolute minimum of f on $[-2, 2]$. A graph of f (see figure 4.11 in your textbook) shows that the critical point $x = 0$ corresponds to neither a local maximum nor a local minimum.

Section 4.2

2 pts

1. Find an interval $I = [a, b]$ on which Rolle's Theorem applies to $f(x) = x^3 - 4x^2 + 4x$. Then find the point(s) c in (a, b) at which $f'(c) = 0$.

Ⓐ $I = [0, 2]$, $c = 2$ and $c = \frac{2}{3}$

Ⓑ $I = [0, 2]$, $c = \frac{2}{3}$

Ⓒ $I = [-2, 2]$, $c = -2$ and $c = 2$

Ⓓ $I = [-2, 2]$, $c = \frac{2}{3}$

Because f is a polynomial, it is everywhere continuous and differentiable. We need an interval $[a, b]$ with the property that $f(a) = f(b)$. Noting that $f(x) = x(x-2)^2$, we choose the interval $[0, 2]$, because $f(0) = f(2) = 0$ (notice, as $x = 2$ is a root with multiplicity 2, no other intervals are possible). The goal is to find points c in the interval $(0, 2)$ at which $f'(c) = 0$, which amounts to the familiar task of finding the critical points of f . The critical points satisfy

$$f'(x) = 3x^2 - 8x + 4 = 0.$$

Using the quadratic formula, the roots are

$$x = \frac{8 \pm \sqrt{8^2 - 4(3)(4)}}{2(3)}, \text{ or } x = \frac{2}{3} \text{ and } x = 2.$$

Hence, the graph of f has exactly one point in the interval $(0, 2)$ at which the tangent line is horizontal. That is, $f(0) = f(2) = 0$, the point $x = 2/3$ is in the interval $(0, 2)$, and $f'(2/3) = 0$.

Section 4.3

2 pts

1. Use the Second Derivative Test to locate the local extrema of the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$ on $(-2, 2)$.
- (A) $f(x)$ has a local minimum at $x = -1$ and a local maximum at $x = 1$.
- (B) $f(x)$ has a local minimum at $x = -1$ and no local maximum.
- (C) $f(x)$ has no local extrema on $(-2, 2)$.
- (D) $f(x)$ has a local minimum at $x = 2$ and a local maximum at $x = -2$.

This function was considered in Examples 6 and 8 in your notes, where we found that $f'(x) = 12(x+1)(x-1)^2$ and $f''(x) = 12(x-1)(3x+1)$. Therefore, the critical points of f are $x = -1$ and $x = 1$. Evaluating f'' at the critical points, we find that $f''(-1) = 48 > 0$. By the Second Derivative Test, f has a local minimum at $x = -1$. At the other critical point, $f''(1) = 0$, so the test is inconclusive. However, you can check that the first derivative does not change sign at $x = 1$, which means f does not have a local maximum or minimum at $x = 1$.

3 pts

2. Identify the intervals on which the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$ is concave up or concave down. Then, locate the inflection points.

- (A) $f(x)$ is concave up on $(1, \infty)$, concave down on $(-\infty, 1)$, and has an inflection point at $x = 1$.
- (B) $f(x)$ is concave up on $(-\infty, -1/3) \cup (1, \infty)$, concave down on $(-1/3, 1)$, and has inflection points at $x = -1/3$ and $x = 1$.
- (C) $f(x)$ is concave up on $(-\infty, -1/3)$, concave down on $(-1/3, \infty)$, and has an inflection point at $x = -1/3$.
- (D) $f(x)$ is concave up on $(-\infty, \infty)$ and has no inflection points.

This function was considered in Example 6, where we found that $f'(x) = 12(x+1)(x-1)^2$. It follows that

$$f''(x) = 12(x-1)(3x+1).$$

We see that $f''(x) = 0$ when $x = 1$ and when $x = -1/3$. These points are candidates for inflection points; to be certain that they are inflection points, we must determine whether the concavity changes at these points.

Testing points, we find that $f''(-1) > 0$ and $f''(2) > 0$. So, f is concave up on the intervals $(-\infty, -1/3)$ and $(1, \infty)$. Similarly, $f''(0) < 0$ and f is concave down on the interval $(-1/3, 1)$.

We can see from the above that the sign of f'' changes at $x = -1/3$ and at $x = 1$, so the concavity of f also changes at these points. Therefore, inflection points occur at $x = -1/3$ and $x = 1$.